Exercise 1

Consider the Navier-Stokes problem defined in the unit square $\Omega := (0,1)^2$ and time interval $I = (0,1)$ by

$$\begin{cases} \dfrac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} - \dfrac{1}{\mathrm{Re}}\Delta \mathbf{u} + \nabla p = \mathbf{f}, & \text{in } \Omega \times I, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega \times I, \\ \mathbf{u} = \mathbf{g}, & \text{on } \partial\Omega \times I, \\ \mathbf{u}(\cdot, t = 0) = \mathbf{u}_0, & \text{in } \Omega, \end{cases}$$

where the forcing $\mathbf{f}$ is given by

$$\mathbf{f}(x, y, t) = \begin{bmatrix} -2\cos x \sin y \left(\cos(2t) + \dfrac{1}{\mathrm{Re}} \sin(2t)\right) \\ 2 \sin x \cos y \left(\cos(2t) + \dfrac{1}{\mathrm{Re}} \sin(2t)\right) \end{bmatrix},$$

and the exact solution $(\mathbf{u}_{\mathrm{ex}}, p_{\mathrm{ex}})$ is defined by

$$\mathbf{u}_{\mathrm{ex}} = \begin{bmatrix} -\cos x \sin y \sin(2t) \\ \sin x \cos y \sin(2t) \end{bmatrix}, \qquad p_{\mathrm{ex}} = -\frac{\cos(2x) + \cos(2y)}{4} \sin^2(2t).$$

1. Solve the problem using the following time-advancing schemes:

   - backward Euler explicit scheme:

   $$\begin{cases} \dfrac{\mathbf{u}_{n+1} - \mathbf{u}_n}{\Delta t} + (\mathbf{u}_n \cdot \nabla)\mathbf{u}_n - \dfrac{1}{\mathrm{Re}}\Delta \mathbf{u}_{n+1} + \nabla p_{n+1} = \mathbf{f}_{n+1}, \\ \operatorname{div} \mathbf{u}_{n+1} = 0, \\ \mathbf{u}_{n+1} = \mathbf{g}_{n+1}; \end{cases}$$

   - backward Euler semi-implicit scheme:

   $$\begin{cases} \dfrac{\mathbf{u}_{n+1} - \mathbf{u}_n}{\Delta t} + (\mathbf{u}_n \cdot \nabla)\mathbf{u}_{n+1} - \dfrac{1}{\mathrm{Re}}\Delta \mathbf{u}_{n+1} + \nabla p_{n+1} = \mathbf{f}_{n+1}, \\ \operatorname{div} \mathbf{u}_{n+1} = 0, \\ \mathbf{u}_{n+1} = \mathbf{g}_{n+1}; \end{cases}$$

2. Study the convergence and stability of each method solving the problem with a decreasing time step ($\Delta t = 1/2,\ 1/4,\ 1/8,\ 1/16$) and an increasing number of subdivisions for the space discretization ($n = 4,\ 8,\ 16,\ 32$).

3. Verify numerically the convergence rates for each method, use a uniform mesh ($n = 10$) and a decreasing time step ($\Delta t = 1/10,\ 1/20,\ 1/40,\ 1/80,\ 1/160$). Compute the error for the velocity and the pressure in the $L^\infty(0,1,H^1(\Omega))$ and $L^\infty(0,1,L^2(\Omega))$ norms, respectively.

Exercise 2

Let us consider the Navier-Stokes problem defined in the domain $\Omega \times I := (0,1)^2 \times (0,10)$

$$\begin{cases} \dfrac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} - \dfrac{1}{\mathrm{Re}}\Delta \mathbf{u} + \nabla p = 0, & \text{in } \Omega \times I, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega \times I, \\ \mathbf{u} = 1\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma^{\mathrm{up}} \times I = \{0 \le x \le 1,\, y = 1\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \backslash \Gamma^{\mathrm{up}} \times I, \\ \mathbf{u}(\cdot, t = 0) = \mathbf{u}_0, & \text{in } \Omega. \end{cases}$$

The initial velocity $\boldsymbol{u}_0$ is assumed to be the solution of the Stokes problem

$$\begin{cases} -\dfrac{1}{\mathrm{Re}}\Delta \boldsymbol{u}_0 + \nabla p_0 = 0, & \text{in } \Omega, \\ \operatorname{div} \boldsymbol{u}_0 = 0, & \text{in } \Omega, \\ \boldsymbol{u}_0 = 1\boldsymbol{i} + 0\boldsymbol{j}, & \text{on } \Gamma^{\mathrm{up}} = \{0 \le x \le 1,\, y = 1\}, \\ \boldsymbol{u}_0 = \mathbf{0}, & \text{on } \partial\Omega \backslash \Gamma^{\mathrm{up}}. \end{cases}$$

1. Solve the problem using the backward Euler semi-implicit time advancing scheme. For each time step solve the monolithic problem obtained using the stable pair of spaces $\mathbb{P}^2/\mathbb{P}^1$.

2. Let us now introduce the SUPG stabilization for the solution of the linearized problem. Use the following stabilization parameter

$$\delta_K = \frac{h_K}{2\|\mathbf{u}\|}\xi(\mathbb{P}e_K),$$

where the local Péclet number is defined as

$$\mathbb{P}e_K := \frac{\|\mathbf{u}\| h_K}{2}\mathrm{Re},$$

and ξ is the *upwind* function

$$\xi(\theta) = \min(1, \theta).$$

Compare the solutions of the previous point with the solutions obtained with the SUPG stabilization.

3. Compare the solutions obtained using the following discretization for increasing values of Reynolds number ($\mathrm{Re} = 10^2,\, 10^3,\, 10^4$):

   - the pair of spaces $\mathbb{P}^2/\mathbb{P}^1$ without SUPG stabilization;
   - the pair of spaces $\mathbb{P}^2/\mathbb{P}^1$ with SUPG stabilization;
   - the pair of spaces $\mathbb{P}^1/\mathbb{P}^1$ with SUPG stabilization.